

CIVL4340

Wind Engineering

Week 6: Turbulence in the wind, Wind tunnels

Overview

- Turbulence (continued)
 - Covariance and correlation
 - Spectra
- Wind tunnel testing
 - Types of wind tunnel (structural engineering application)
 - Generating an ABL in a wind tunnel
 - Scaling requirements
 - Types of wind tunnel tests

Covariance & Correlation

- Gives information on the temporal and spatial distribution of the wind
 - Large structures - spatial correlation
 - Dynamically sensitive structures – temporal correlation
- Covariance – mean product of the fluctuating components at one or more points either simultaneously or with a time lag

Covariance of
records i & j given
time lag τ

$$C_{ij}(\tau) = \overline{i(t) \cdot j(t + \tau)} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T i(t) \cdot j(t + \tau) dt$$

Fluctuating
component
of record 1

Fluctuating
component
of record 2

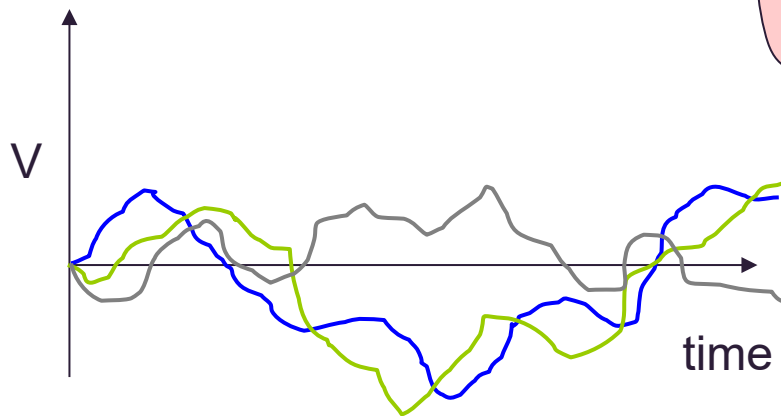
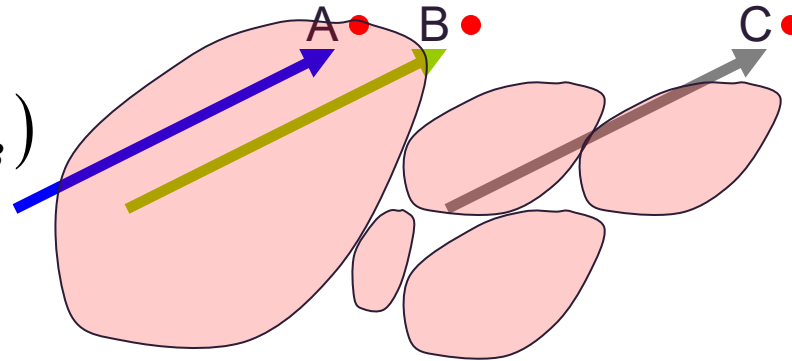
Time lag

Covariance in space

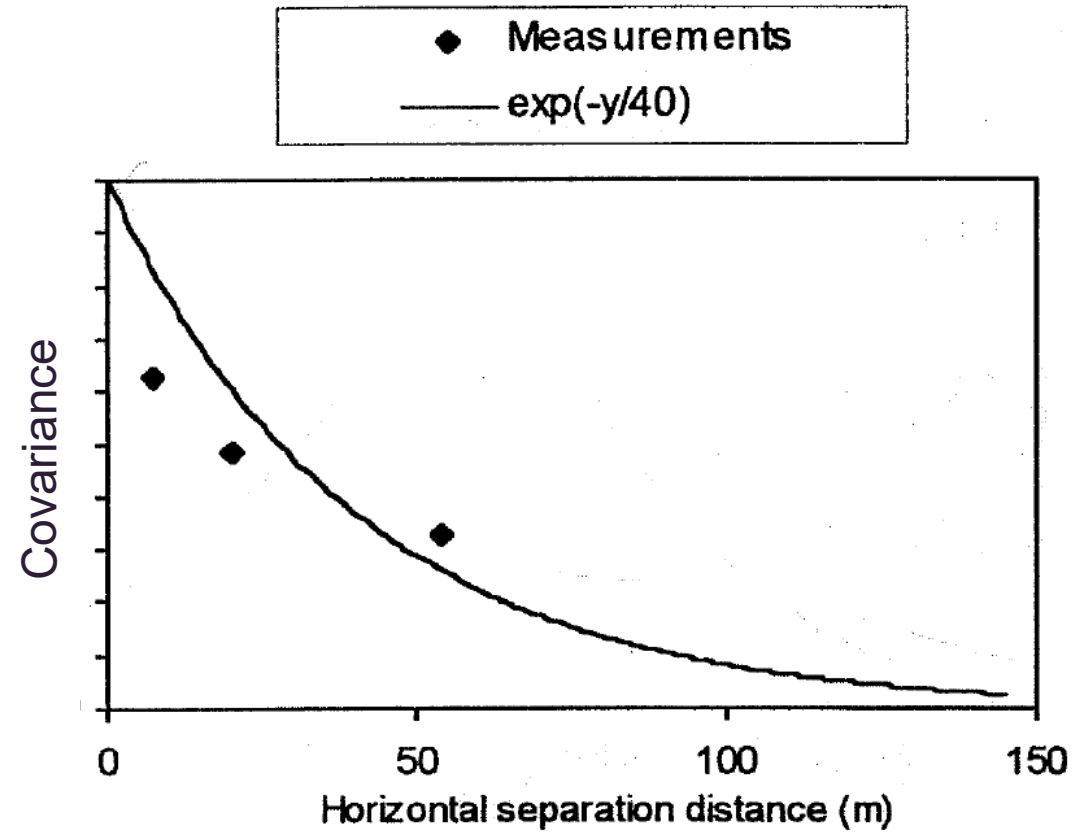
$$C_{ij}(\tau) = \frac{1}{T} \int_0^T i(t) \cdot j(t + \tau) dt$$

$$i(t) = V_A(z_A, t) - \bar{U}(z_A)$$

$$j(t + \tau) = V_B(z_B, t + \tau) - \bar{U}(z_B)$$



Covariance in space



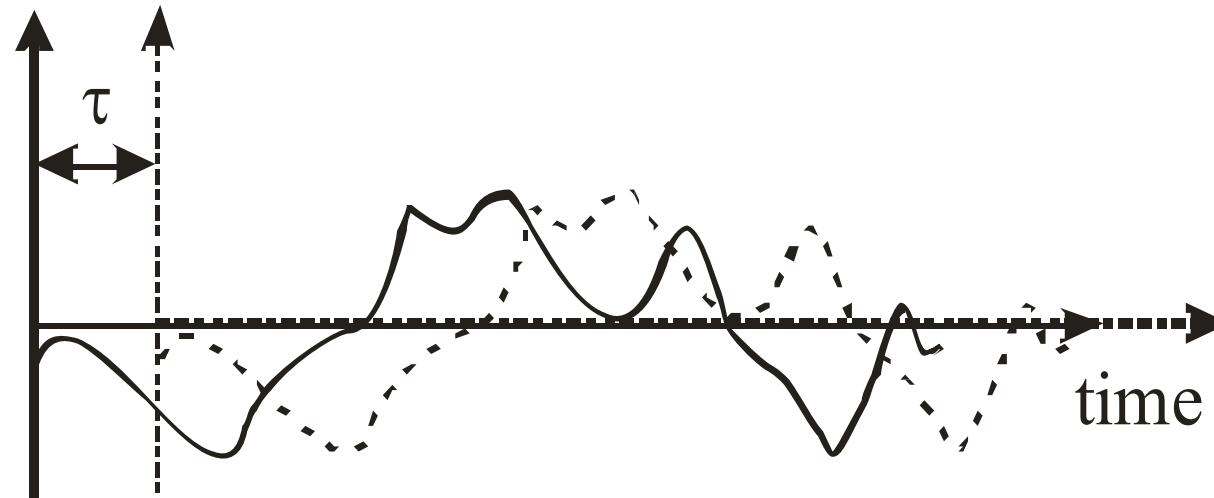
Covariance in time

$$C_{ij}(\tau) = \frac{1}{T} \int_0^T i(t) \cdot j(t + \tau) dt$$

$$i(t) = V_A(z_A, t) - \bar{U}(z_A)$$

$$j(t + \tau) = V_B(z_B, t + \tau) - \bar{U}(z_B)$$

Fluctuating
wind speed



Covariance

- Measurements at a single point – autocovariance.
 - Temporal information.
- Measurements at multiple points – covariance.
 - Temporal and spatial information.
- Nine possible covariance functions.
 - Important: autocovariance, uv , and uw .
- At $\tau = 0$, autocovariance = variance, therefore useful to normalise using variance.

Correlation

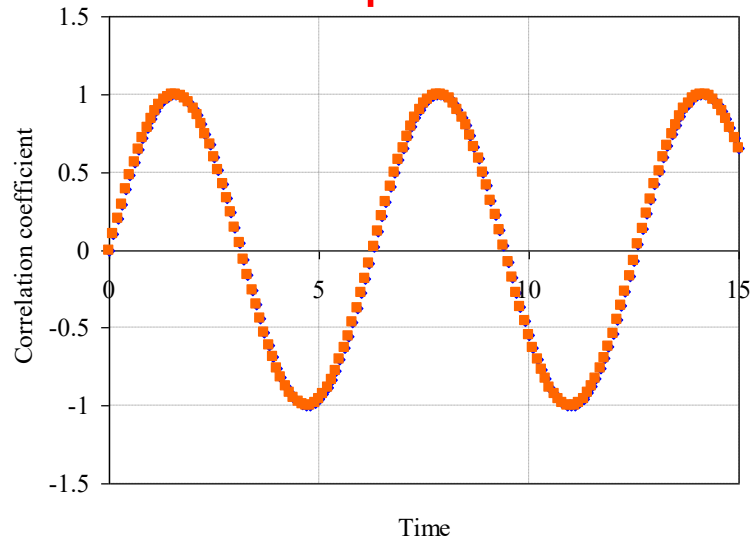
- Correlation coefficient (normalised covariance)

Correlation coefficient $\rightarrow \rho_{ij} = \frac{C_{ij}}{\sigma_i \cdot \sigma_j}$ $-1 \leq \rho \leq 1$

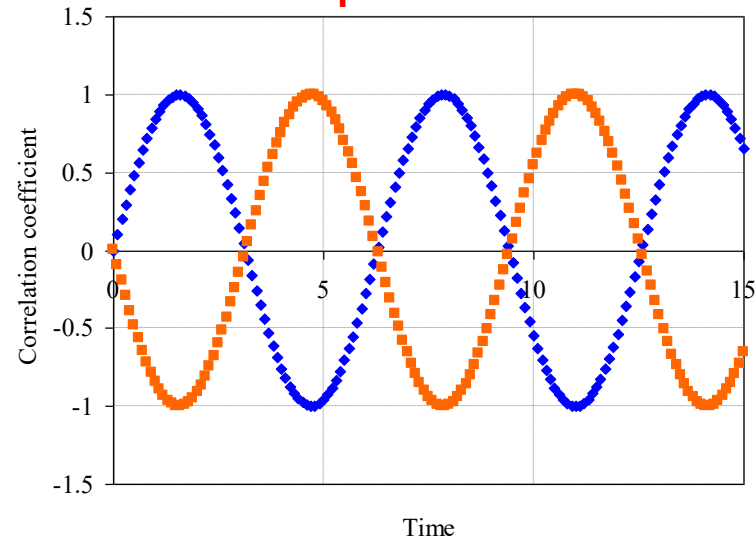
- Autocorrelation high when τ is small
- As τ increases, autocorrelation decreases
- Cross-correlation high when spacing between measurement locations is small and decreases with spacing

Correlation

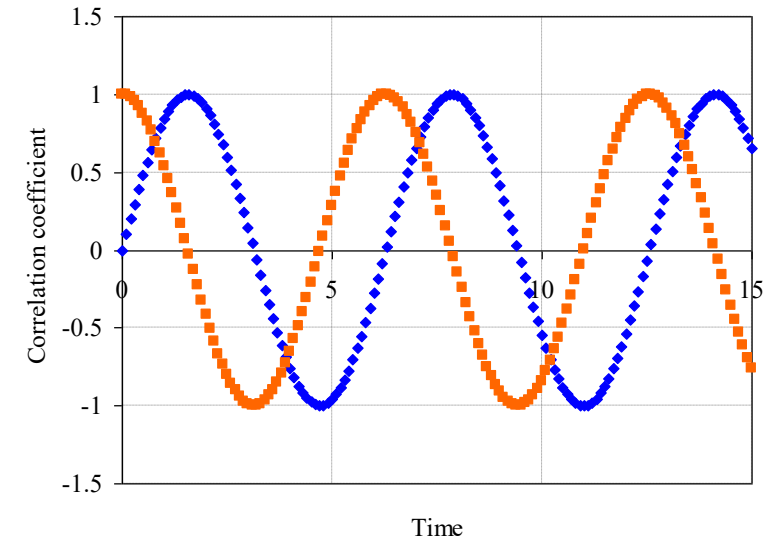
$$\rho = 1$$



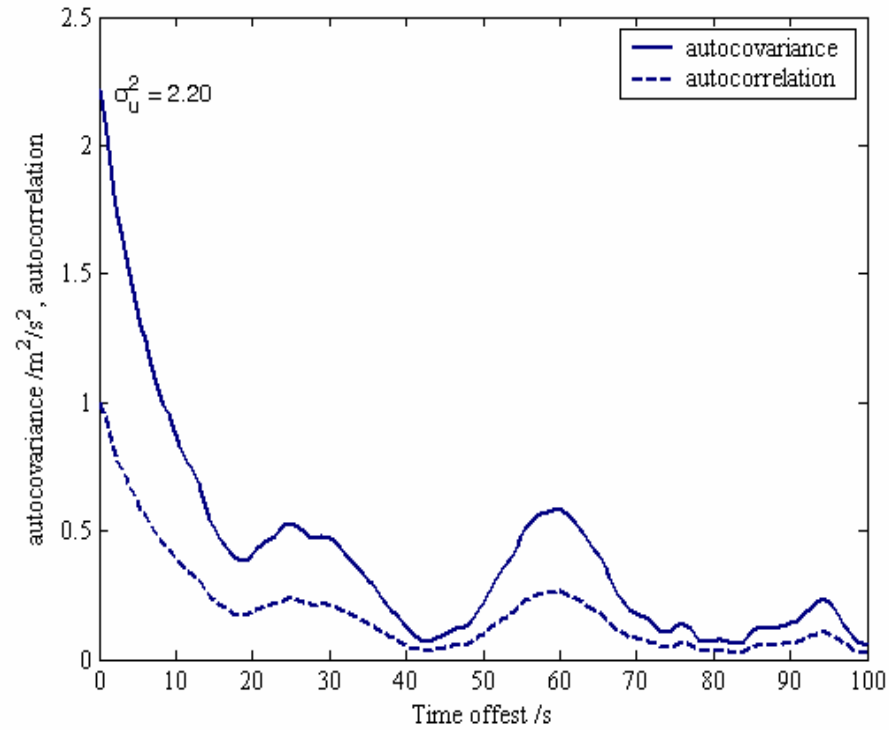
$$\rho = -1$$



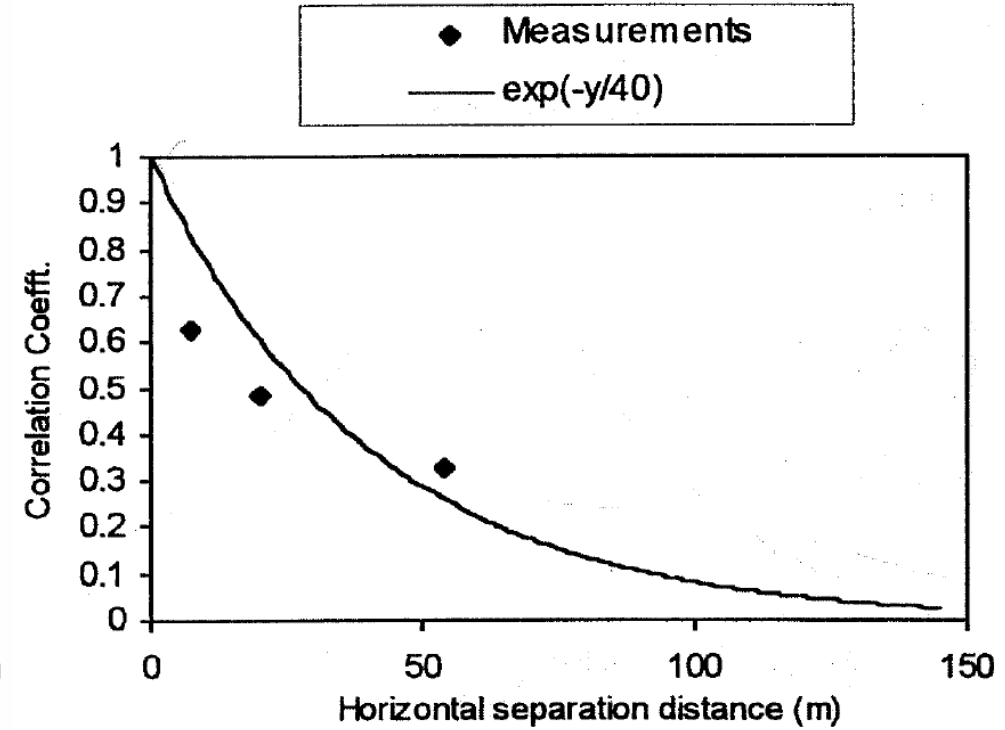
$$\rho = 0$$



Example data

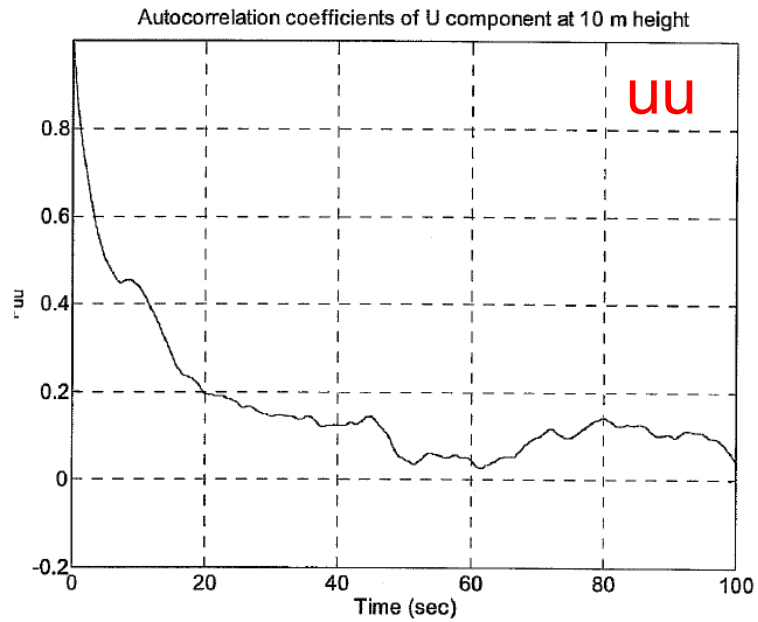


Temporal

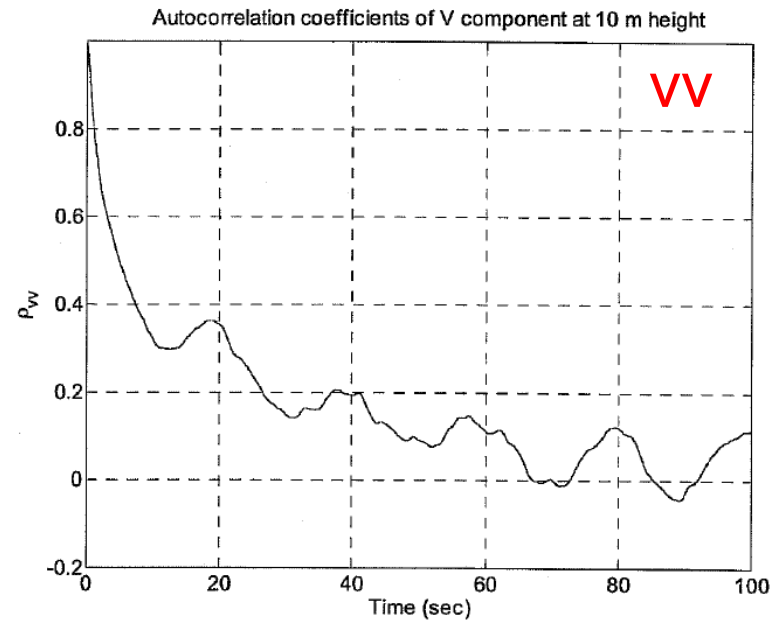


Spatial

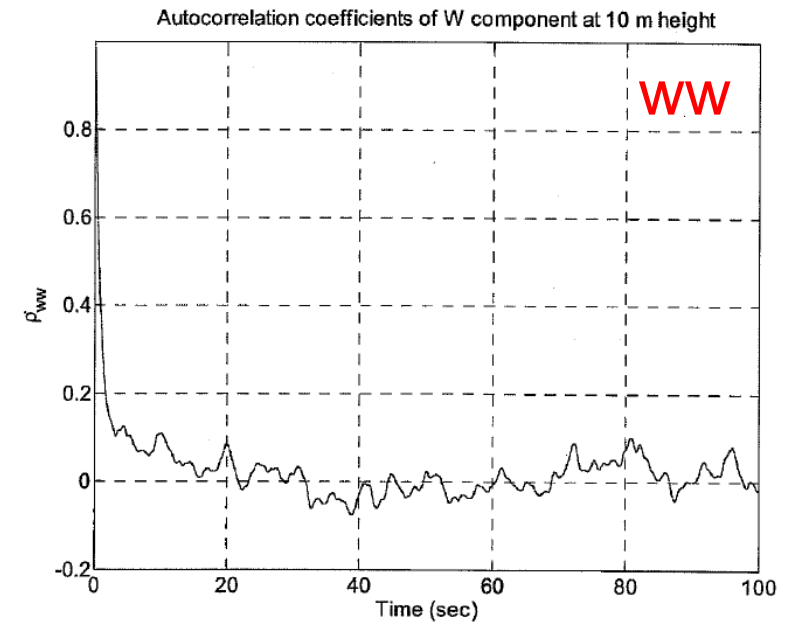
Example data



(a) U component



(b) V component



(c) W component

Gust size

Using auto-correlation information we can define the duration and size of a *typical* gust

Time scale

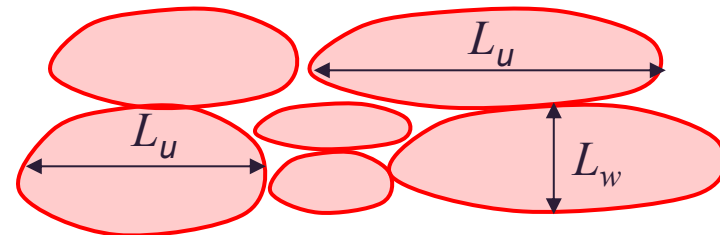
$$T_i = \int_0^{\infty} \rho_{ii}(\tau) d\tau$$

$\rho_{uu}, \rho_{vv}, \rho_{ww}$

(Note: In the original image, the term ρ_{ii} in the integral is circled in red, and a red arrow points from the text $\rho_{uu}, \rho_{vv}, \rho_{ww}$ to this circle.)

Length scale

$$L_i = T_i \bar{U}$$



Gust size

Length scale L_u

$$L_u = \int_{-\infty}^{\infty} \rho_{ij}(\tau = 0, s) ds$$

Cross correlation coefficient

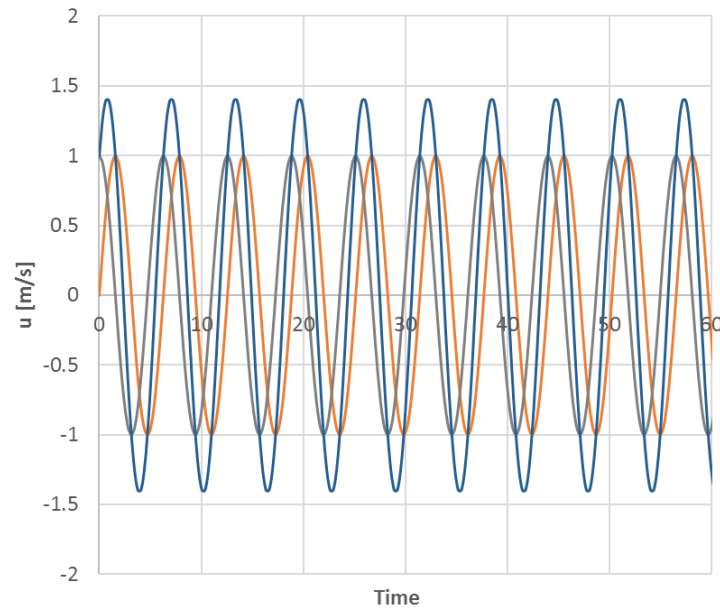


Table 1 Summary of 10 m height wind speed data in Fig. 2. Mean velocity = 12.3 m/s.

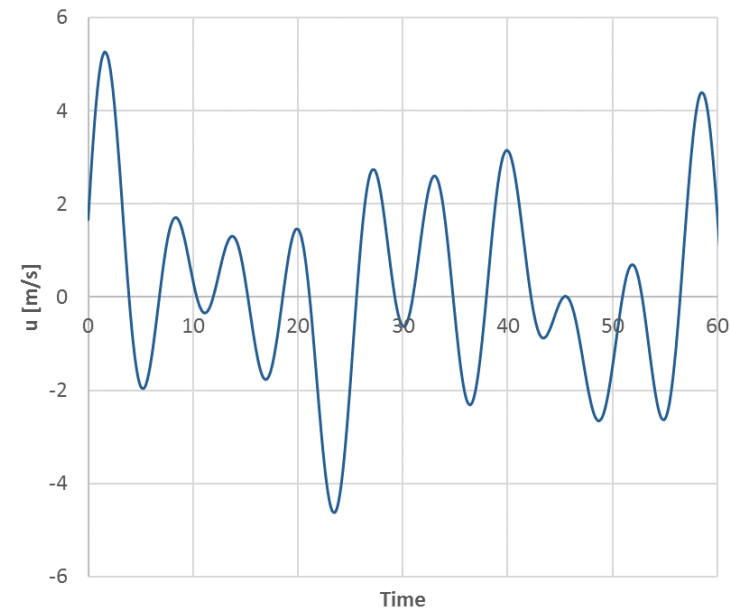
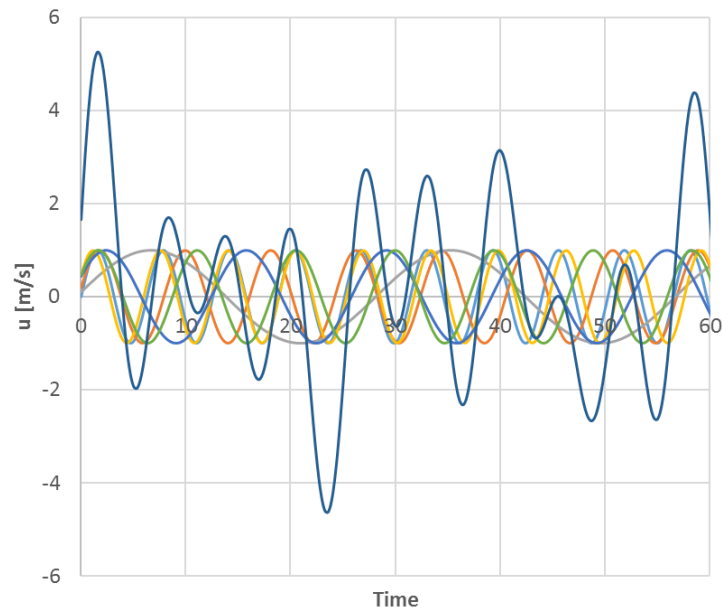
Component i	I_i	T_i (s)	L_i (m)
u	0.144	17.4	213
v	0.130	15.8	193
w	0.068	2.38	27.4

Turbulence - spectra

- Assuming a stationary random process, we assume any signal can be made up of sine and cosine curves.
- In wind, this is the superposition of different sized eddies with a circular frequency of $\omega = 2\pi n$.



Turbulence - spectra



Turbulence - spectra

- Power spectrum requires Fourier Transformation of the time series or the (auto)covariance function.

Spectral density

Fourier Transform

$$S_x(n) = \lim_{T \rightarrow \infty} \left[\frac{2}{T} |X_T(n)|^2 \right]$$

Time

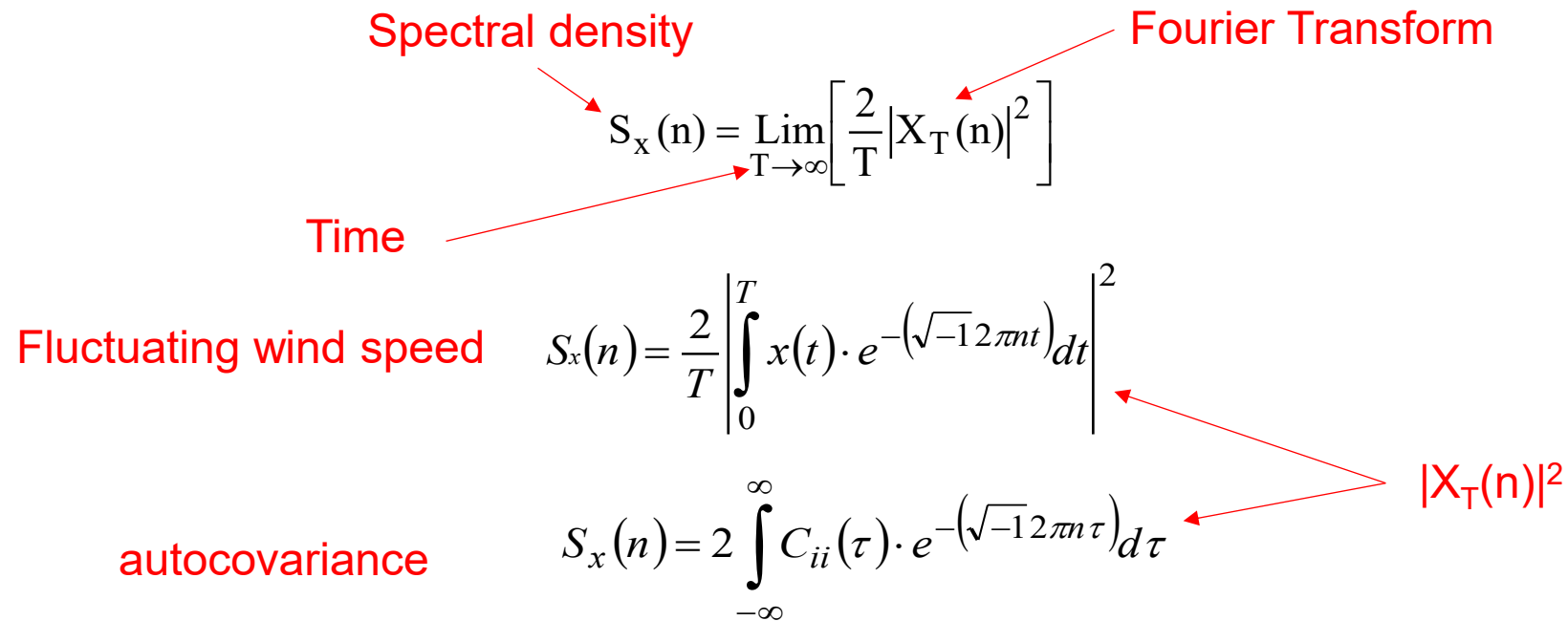
Fluctuating wind speed

$$S_x(n) = \frac{2}{T} \left| \int_0^T x(t) \cdot e^{-\sqrt{-1} 2\pi n t} dt \right|^2$$

autocovariance

$$S_x(n) = 2 \int_{-\infty}^{\infty} C_{ii}(\tau) \cdot e^{-\sqrt{-1} 2\pi n \tau} d\tau$$

$|X_T(n)|^2$



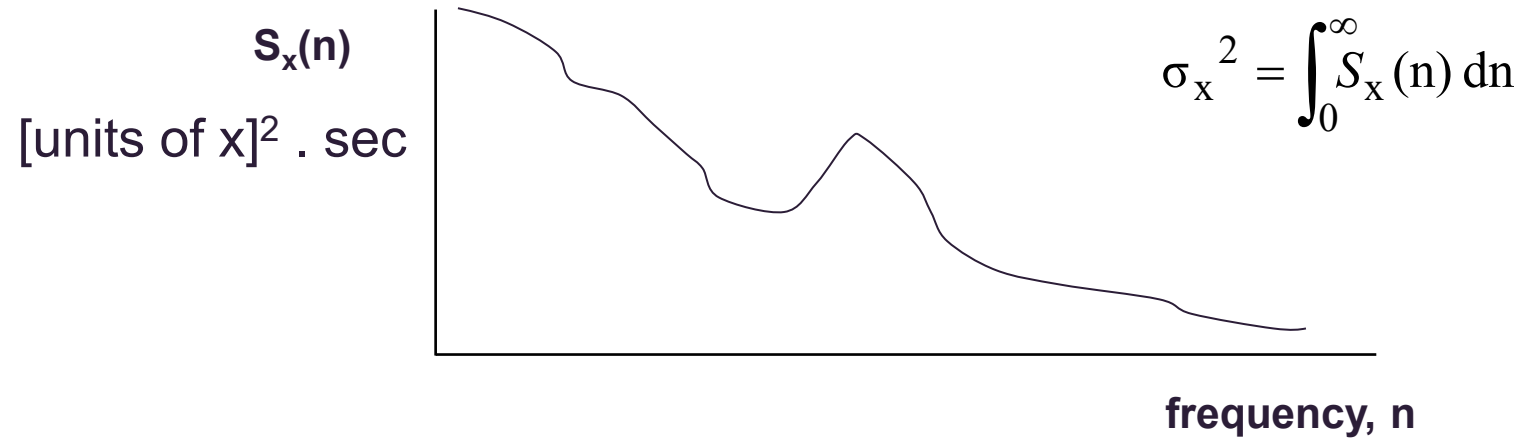
The diagram illustrates the mathematical relationships between different representations of turbulence spectra. At the top, 'Spectral density' points to the limit equation for $S_x(n)$. 'Fourier Transform' points to the $|X_T(n)|^2$ term in the same equation. 'Time' points to the T in the denominator of the limit equation. 'Fluctuating wind speed' points to the $x(t)$ term in the integral equation for $S_x(n)$. 'autocovariance' points to the $C_{ii}(\tau)$ term in the integral equation for $S_x(n)$. Finally, $|X_T(n)|^2$ is shown on the right, with red arrows pointing to it from the 'Fourier Transform' label and the 'autocovariance' label, indicating its role as the squared magnitude of the Fourier transform of the time series.

- Fourier Transform is a complex argument where the absolute value describes the frequency content, and the imaginary component describes the phase offset.

Turbulence - spectra

- Fourier Transform usually solved computationally through a Fast Fourier Transform (FFT) procedure.
- Reciprocal is an Inverse FFT, which converts a spectra (frequency domain) back to a time series.

Turbulence - Spectra



- Area under spectra = variance.
- Spectra is often normalised by variance, $n.S_x(n)/\sigma^2$
- Problem: to calculate a spectra you need a full time series of data.

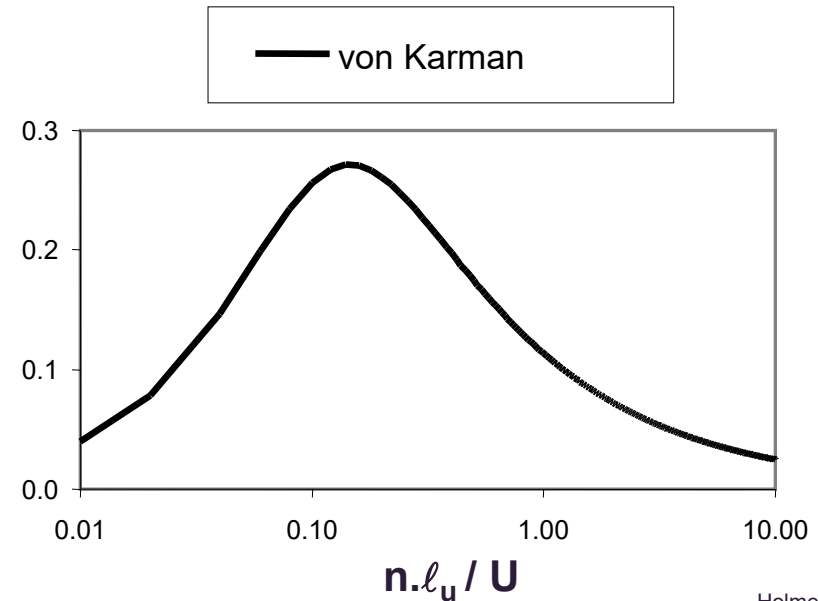
Turbulence - spectra

- Empirical spectra are widely used.
- Many forms, but the von Karman spectra (modified by Harris for wind engineering) is the most widely used for u -component.

Integral length scale

$$\frac{n \cdot S_u(n)}{\sigma_u^2} = \frac{4 \left(\frac{n \cdot \ell_u}{\bar{U}} \right)}{\left[1 + 70.8 \left(\frac{n \cdot \ell_u}{\bar{U}} \right)^2 \right]^{5/6}}$$

$n \cdot S_u(n) / \sigma_u^2$

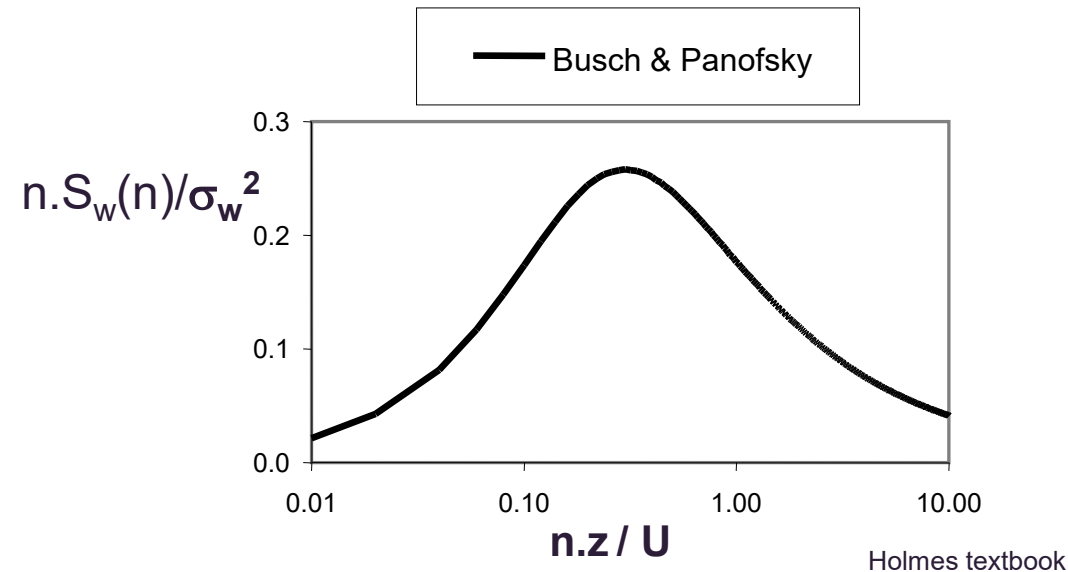


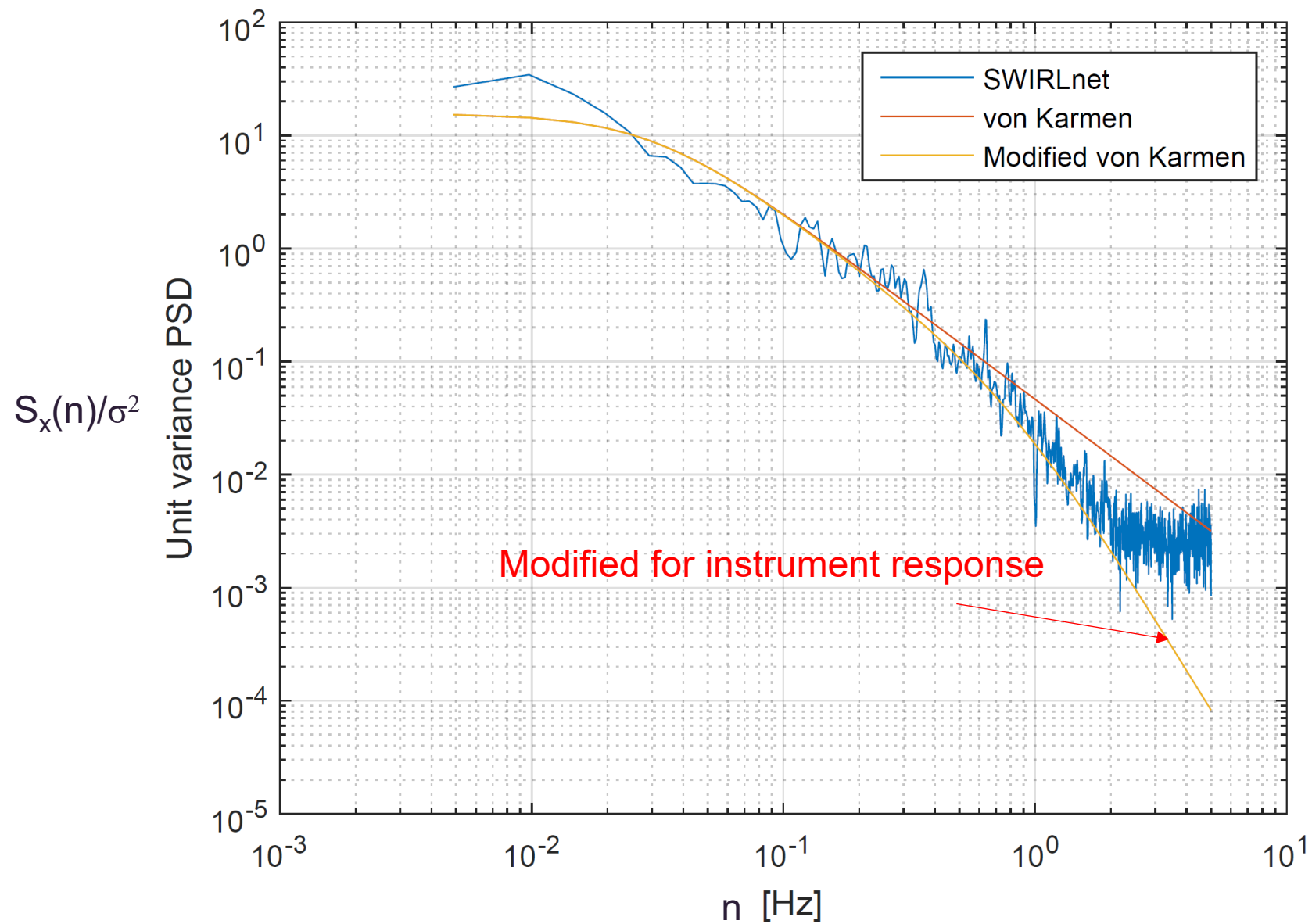
Turbulence - spectra

- Busch and Panofsky for v-component.

$$\frac{n.S_w(n)}{\sigma_w^2} = \frac{2.15 \left(\frac{nz}{\overline{U}} \right)}{\left[1 + 11.16 \left(\frac{nz}{\overline{U}} \right)^{5/3} \right]}$$

elevation





Turbulence quantities - summary

- I_u , G , g describe the magnitude of turbulence.
- L describes the mean size of turbulent eddies.
- For dynamically sensitive structures the frequency of turbulence is important.
- Analysis moved from the time domain to the frequency domain.
- Spectral density (spectra, spectrum, power spectrum) of the wind describes the distribution of turbulence fluctuations with frequency, n .
- More specifically, the spectra describes the contribution to variance of winds in the range n to $n + dn$.

Wind tunnel testing

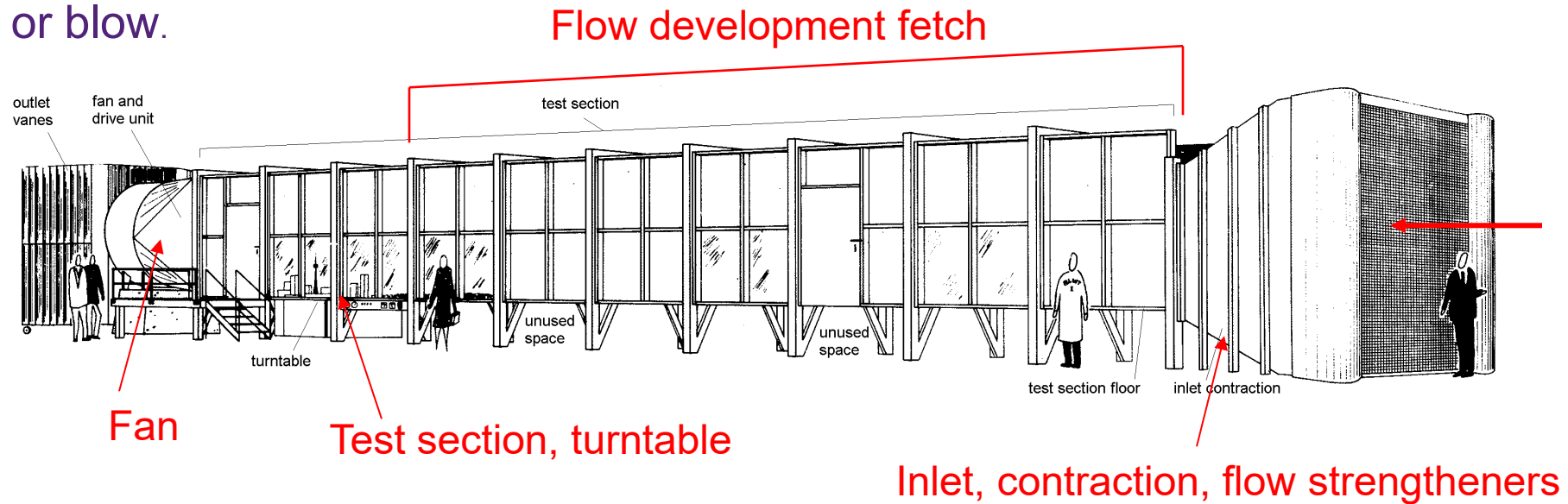
Aim of wind tunnel testing

- Simulate the design wind event appropriate for a particular site for the purposes of predicting design wind loads on, or wind conditions around, structures.
- Generally, only need to simulate the lowest few hundred metres of the ABL.
- Is a standard wind tunnel the most appropriate tool in all cases?
- Computational techniques do not capture turbulent fluctuations acceptably, so do not always generate accurate peak loads – this doesn't mean they can't be used.

Types of wind tunnel

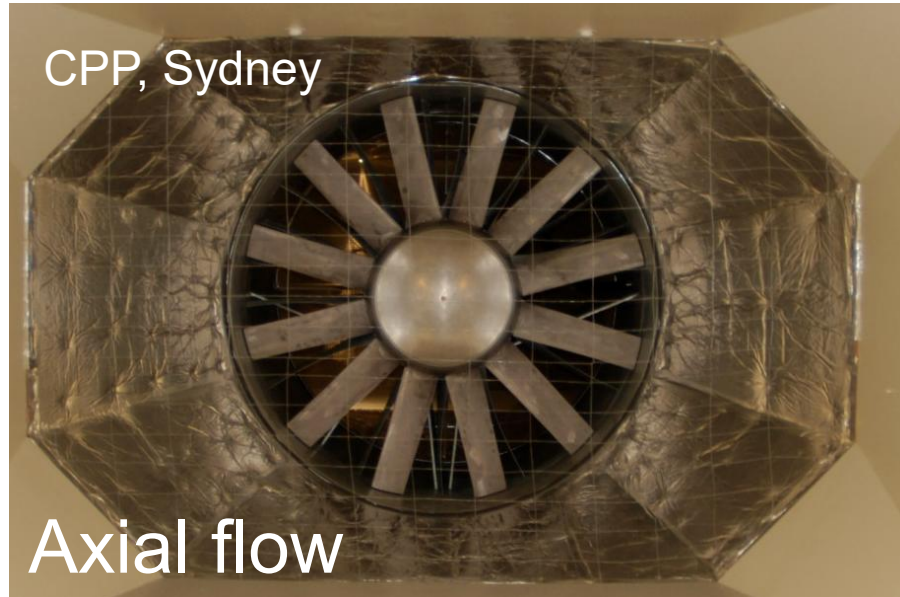
Open circuit wind tunnel

- Flow smoothed at inlet (not essential for BLWT).
- Diffuser at outlet reduces h_L
- Suck or blow.



University of Western Ontario BLWT

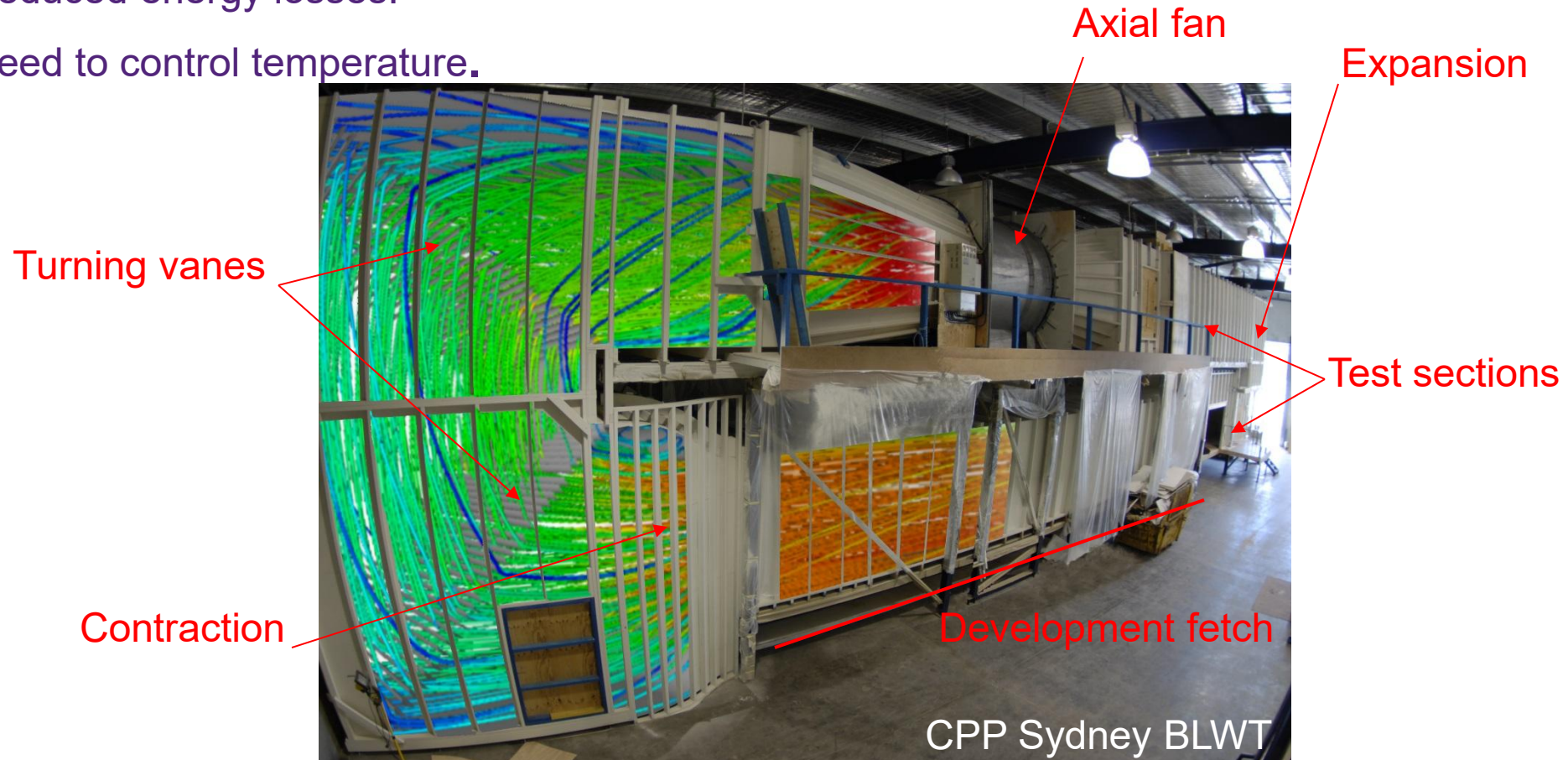
Fans



Types of wind tunnel

Closed circuit wind tunnel

- Can have multiple test sections.
- Reduced energy losses.
- Need to control temperature.



Types of wind tunnel



Mixed type

- E.g. Open circuit tunnel in enclosed building



Swirl reducing
screens needed

Wind tunnel size

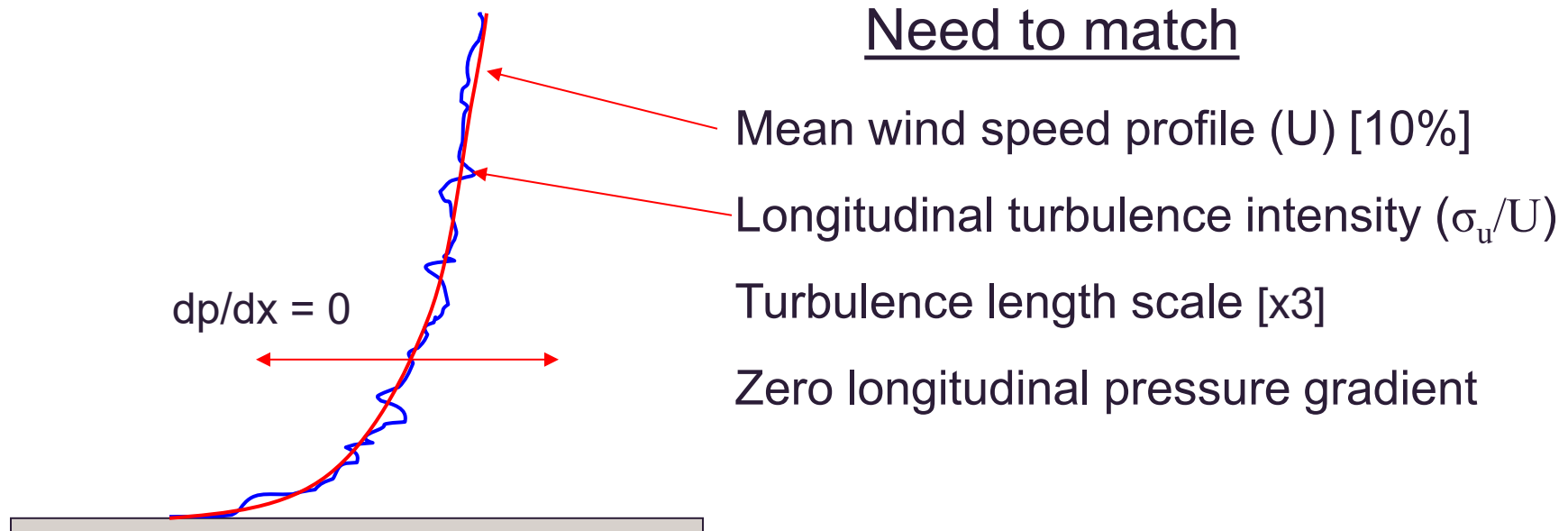
Q: How big does a wind tunnel need to be?

A: depends on what you want to test.



Simulating the natural wind

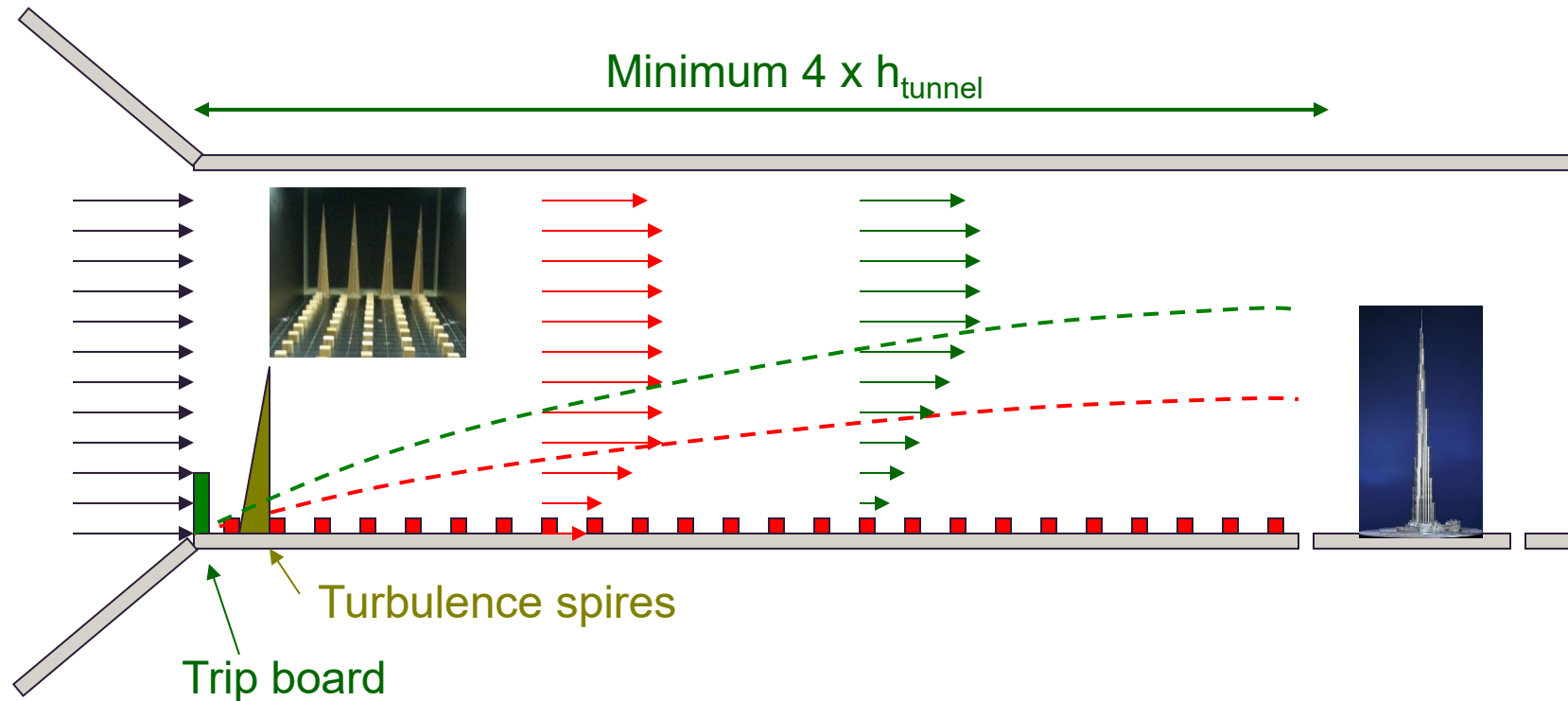
- Wind tunnels simulate “approach” flow



Simulating the natural wind

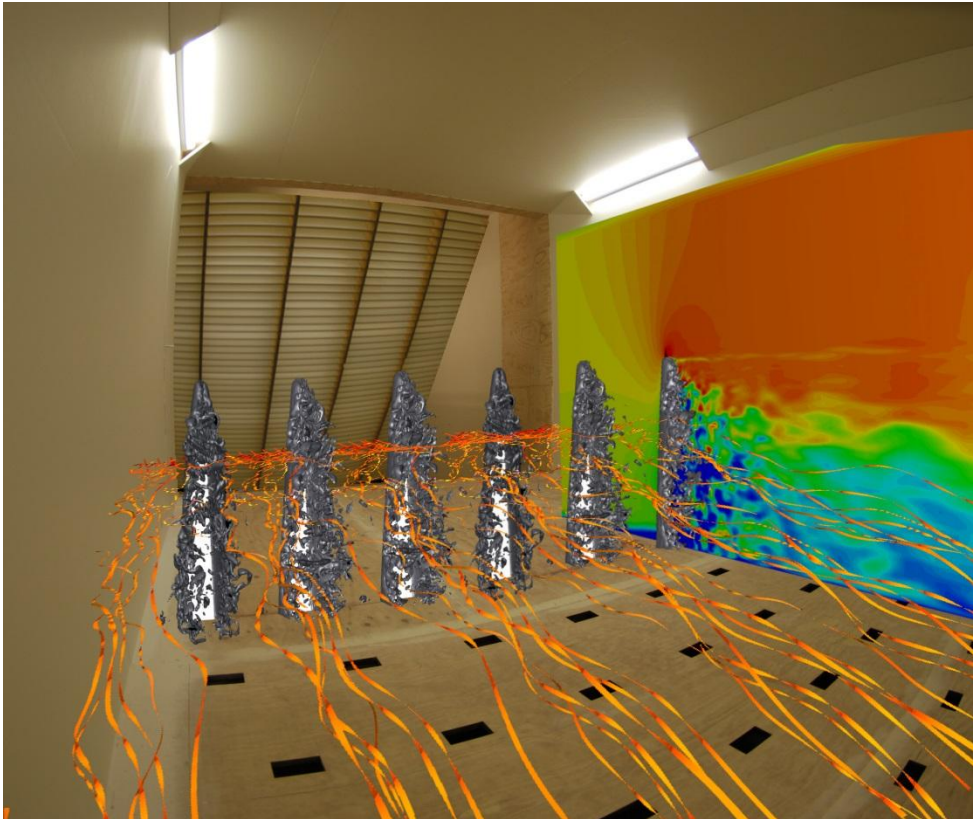
Method depends on available length

Ideally let it develop “naturally”, but may need some help

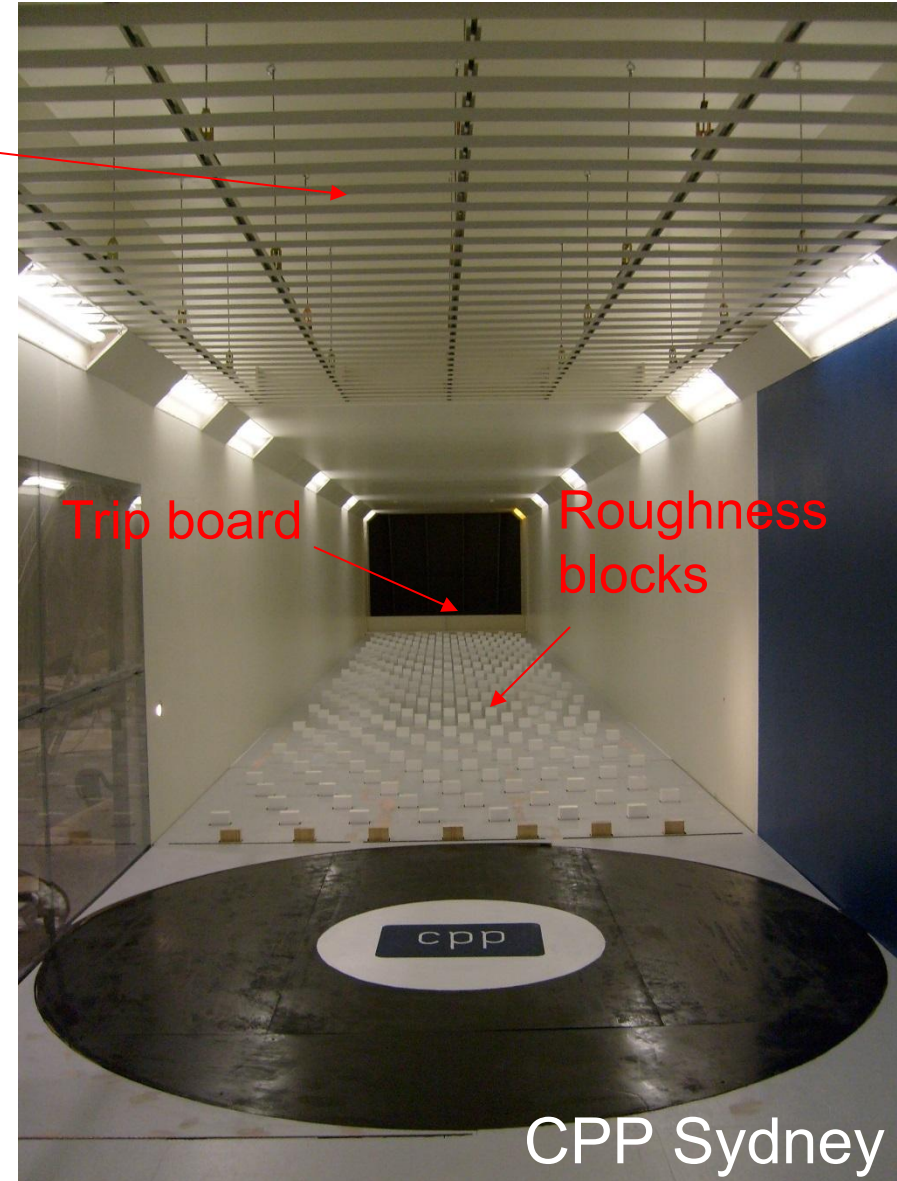


Simulating the natural wind

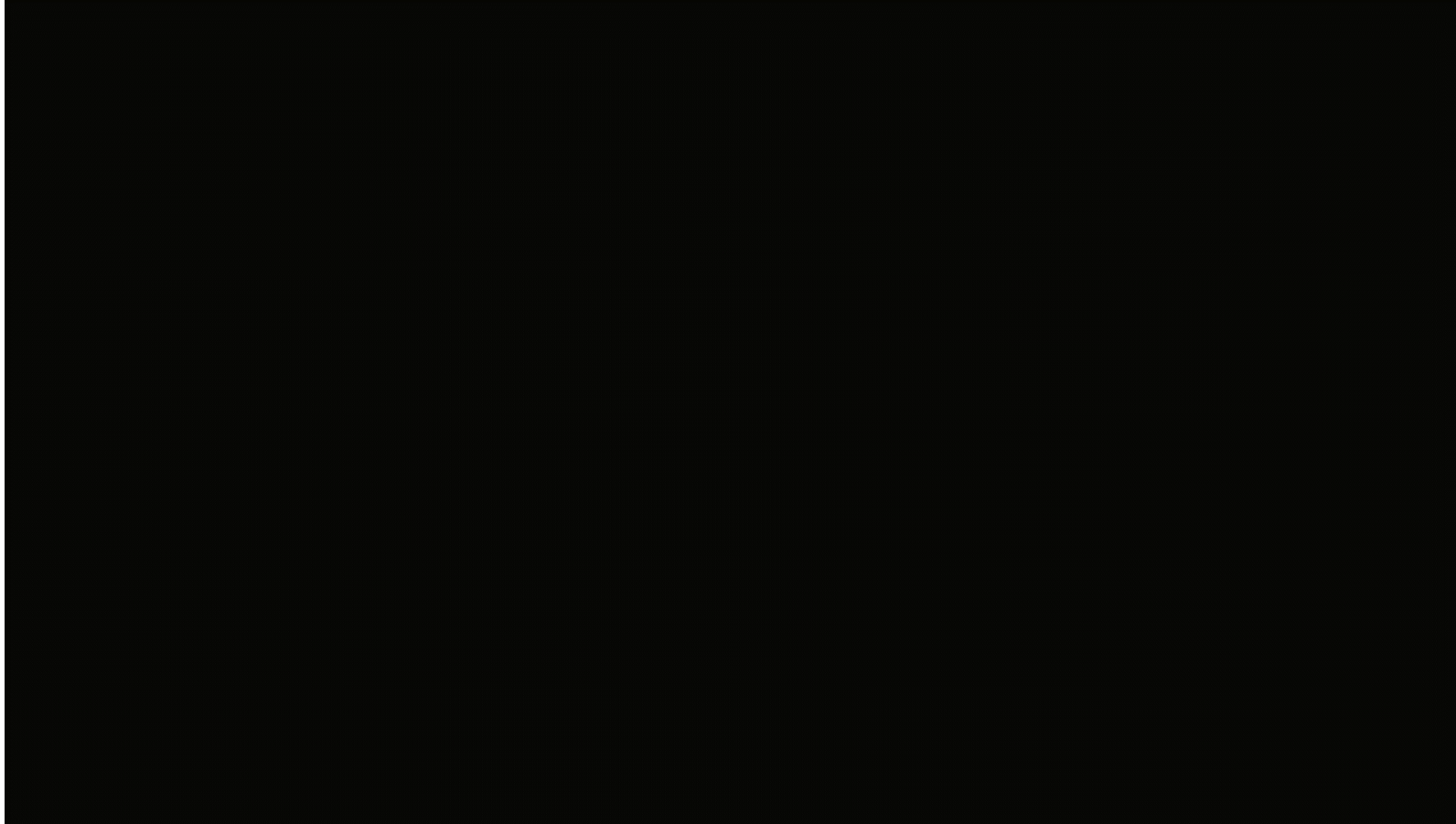
Blockage tolerance
(blockage < 10% can be relaxed to < 30%)



Turbulence spires

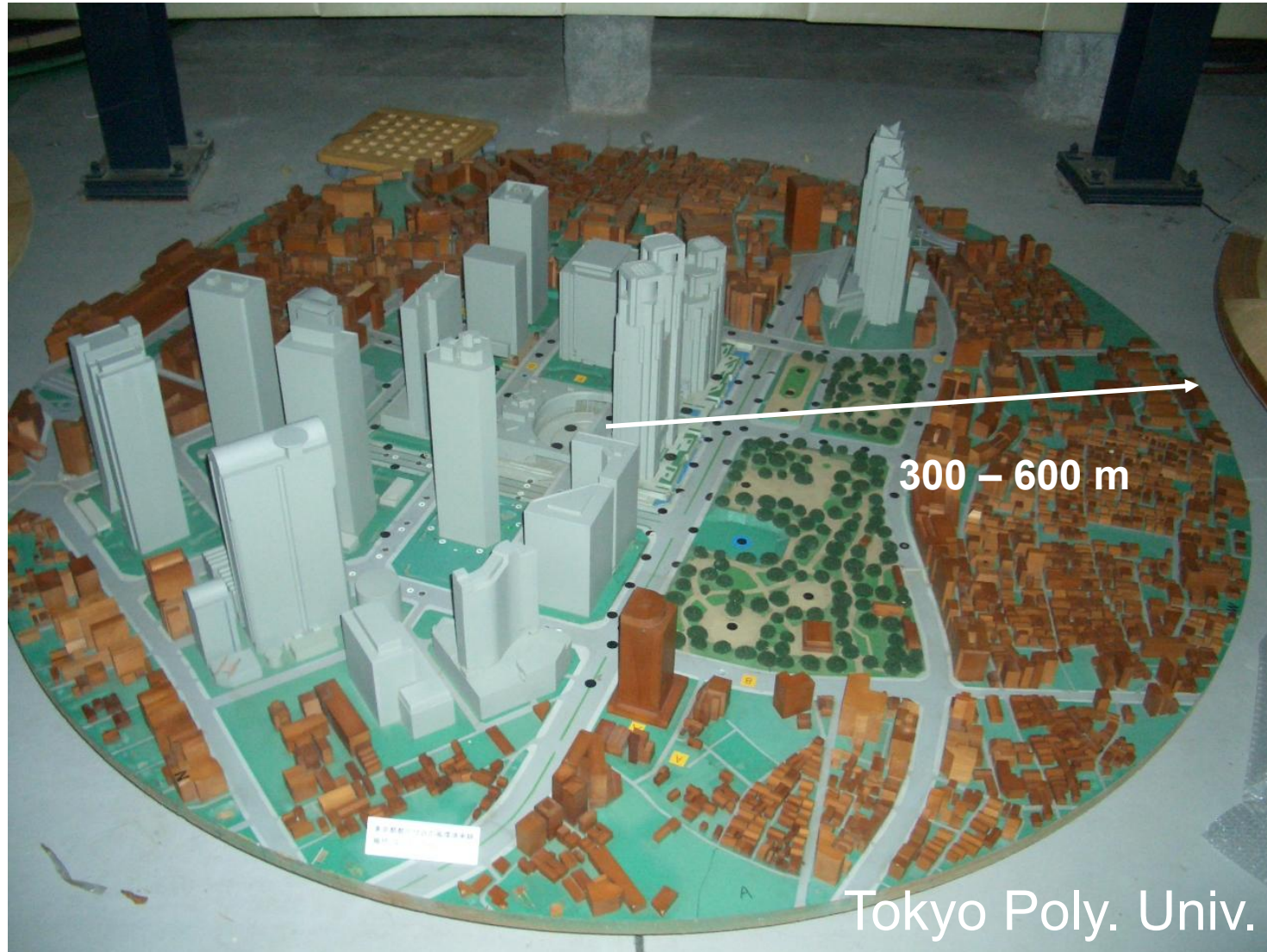


Simulating the natural wind



https://www.youtube.com/watch?v=_VKqqrQkc5k

Simulating the natural wind

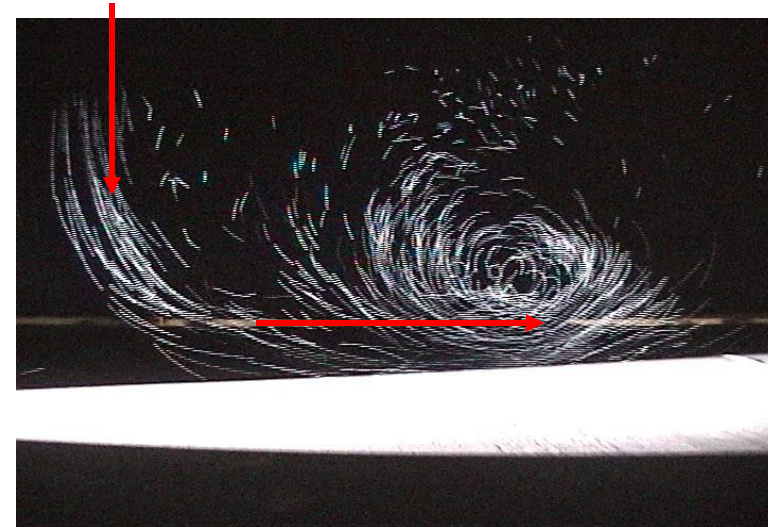


Tropical cyclones

- Structure not known exactly.
- Outer low wind speed regions, “standard” boundary layer seems adequate.
- Peak winds occur just outside eye wall, highly convective, steep mean velocity gradients, high (patchy) turbulence – wind tunnel probably ok, but still an active area of research.

Downburst wind tunnel

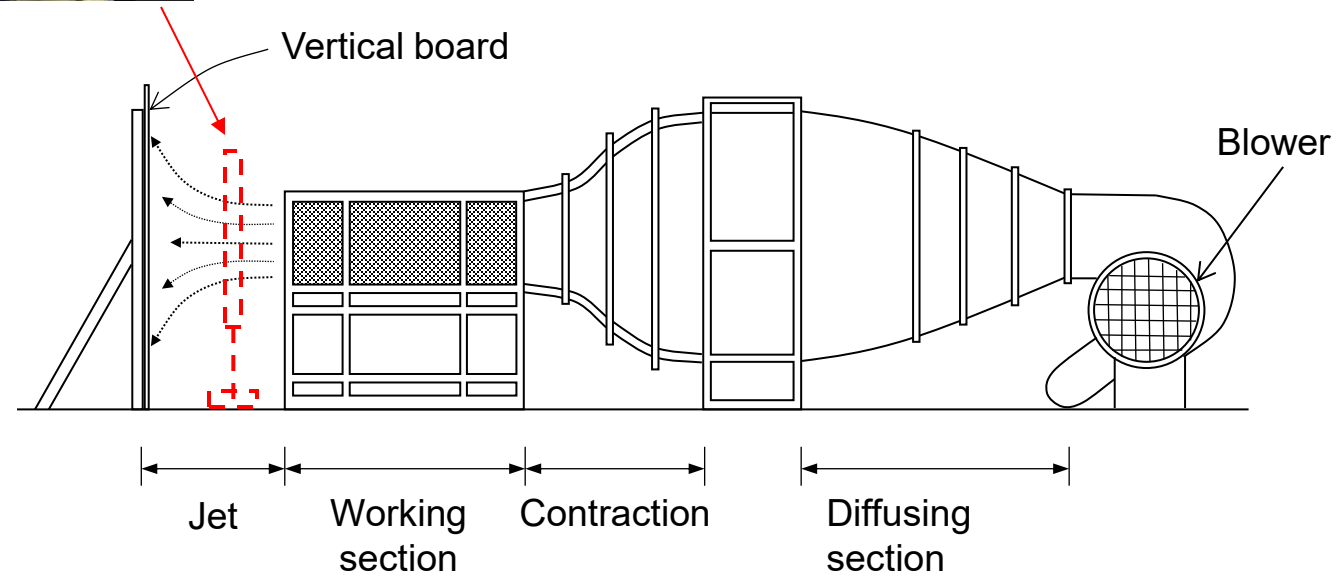
Texas Tech University



Downburst wind tunnel



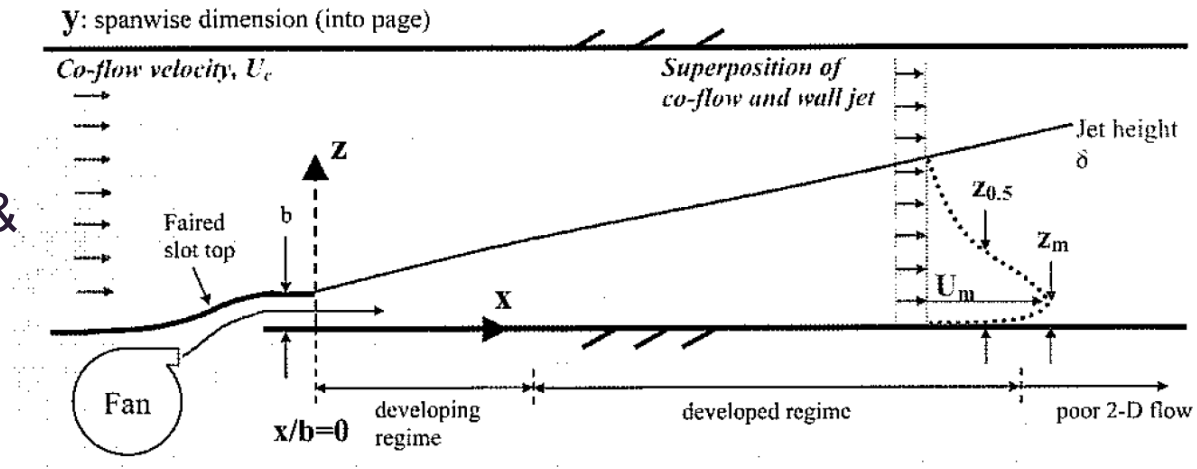
Mechanism for pulsing
the flow (Usyd)



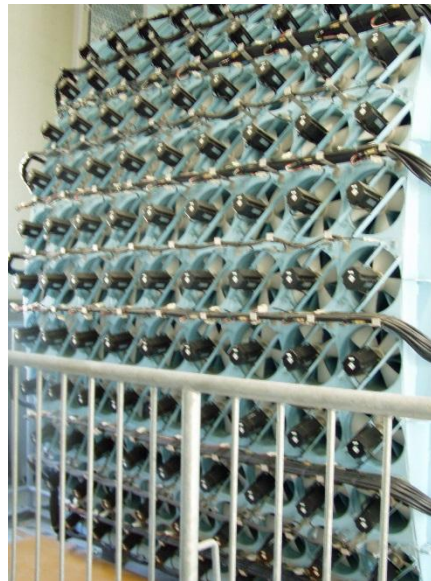
(CSIRO)

Downburst wind tunnel

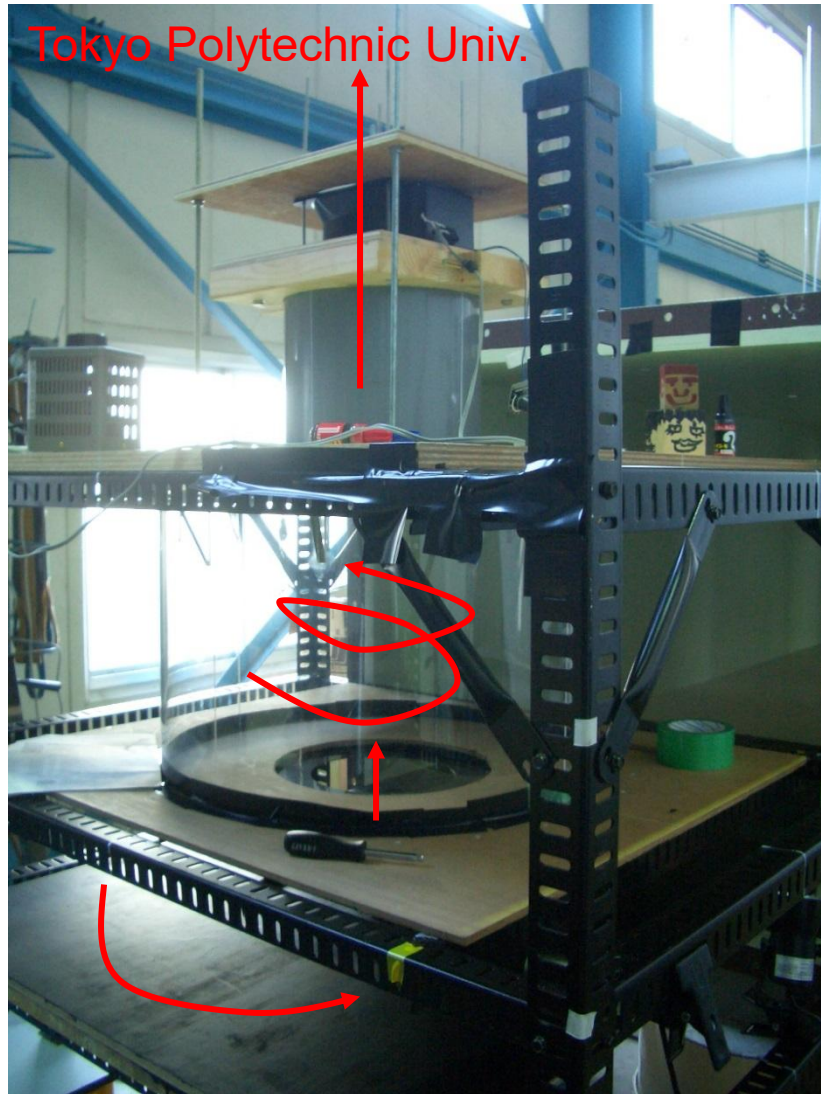
Western
University (Lin &
Savory 2006)



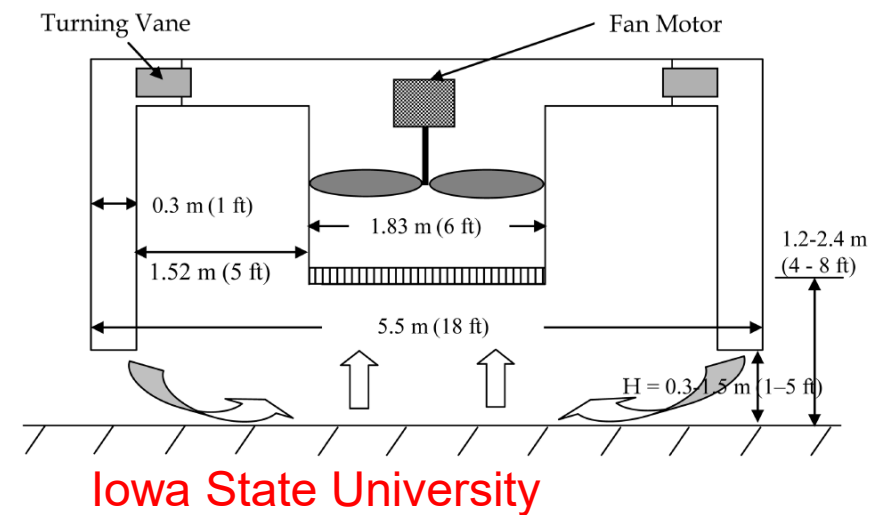
University of
Miyazaki multiple
jet wind tunnel



Tornado simulators

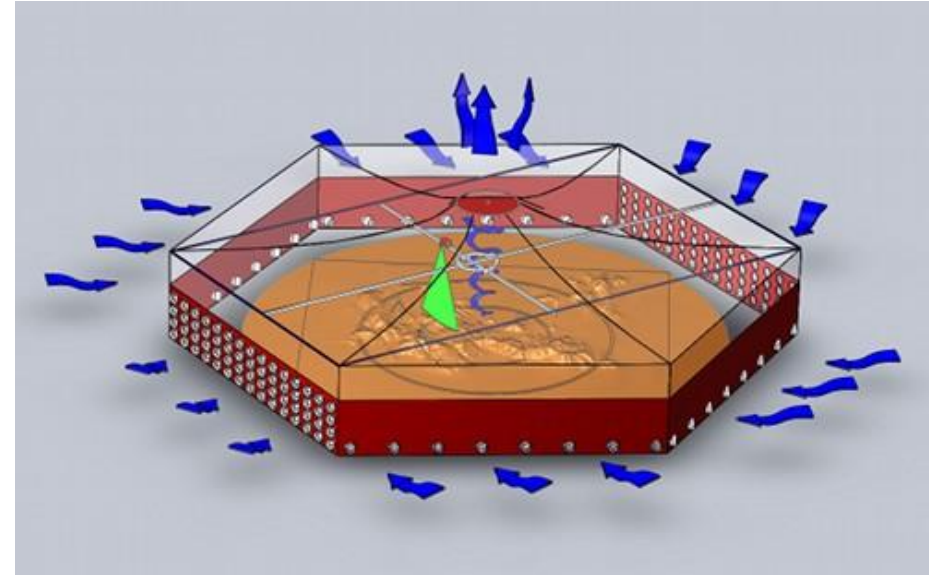


TTU VorTech



Combined facility

Wind Engineering, Energy and
Environment (WindEEE) Dome,
Western University, Canada.



Dimensional analysis

- Not possible to obtain equality of all non-dimensional groups – knowledge of phenomena required.

- Most important

$$L_R = z_{0,m} / z_{0,p} = L_{u,m} / L_{u,p} \quad (\text{Length scale})$$

$$V_R = V_{\text{ref},m} / V_{\text{ref},p} \quad (\text{Velocity scale})$$

$$\text{Time and frequency scaling through } V_R = L_R / T_R$$

$$Re = \rho_m L_m V_m / \mu_m = \rho_p L_p V_p / \mu_p \quad (\text{Re matching – not generally possible})$$

- Ensure $Re > 10^5$ (sharp edged bodies)
- Further requirements when structures start to move!

$$St = nL/V \quad (\text{Strouhal Number})$$

$$Fr = V/(gL)^{1/2} \quad (\text{Froude Number})$$

Wind tunnel tests

Pressure measurements

- Cladding loads
- Overall structural loads
- Fixing design issues

Overall load and response

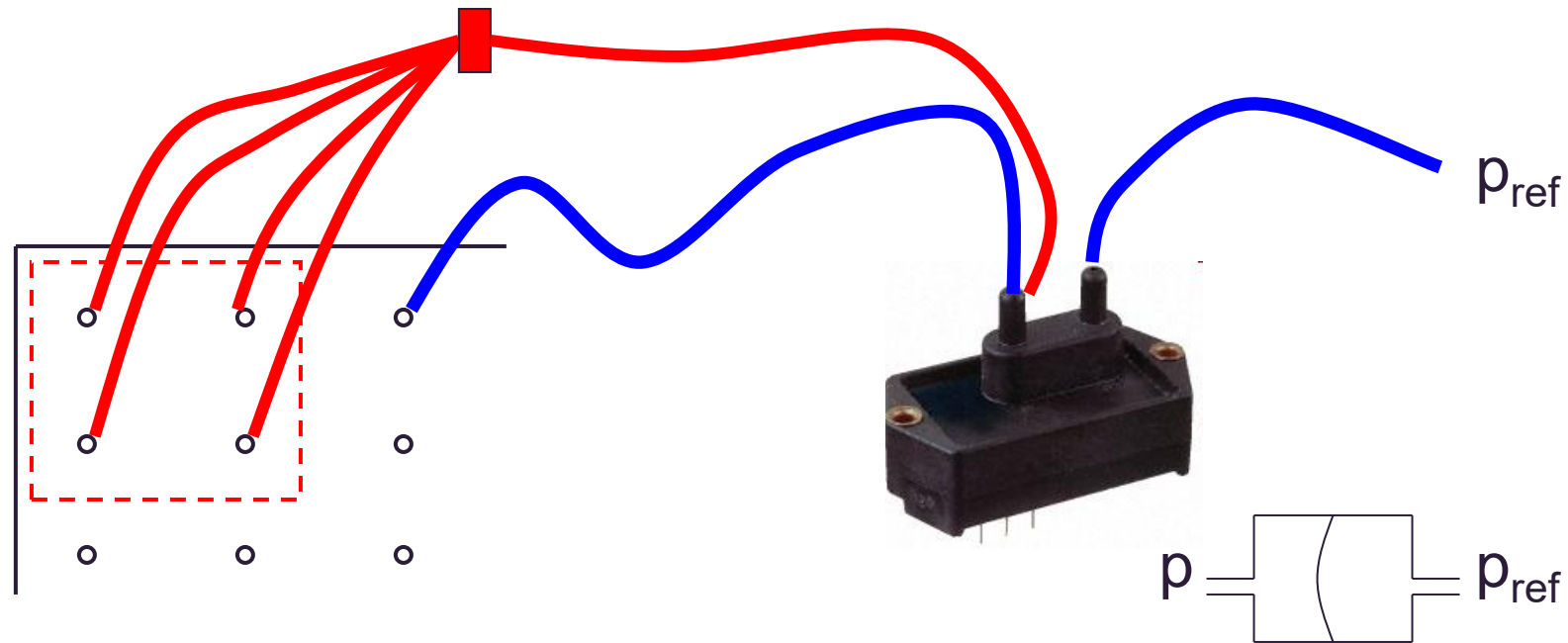
- Base bending and torsion moments and shear
- Prediction of deflection and accelerations

Environmental

- Pedestrian/occupant comfort
- Pedestrian safety
- Pollutant dispersion

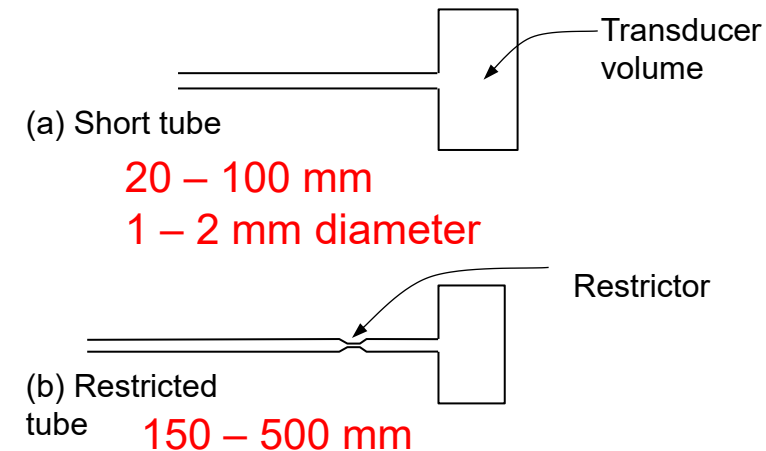
Pressure measurements

- Measurement of fluctuating and peak pressures.
- Single or multiple tap measurements made.

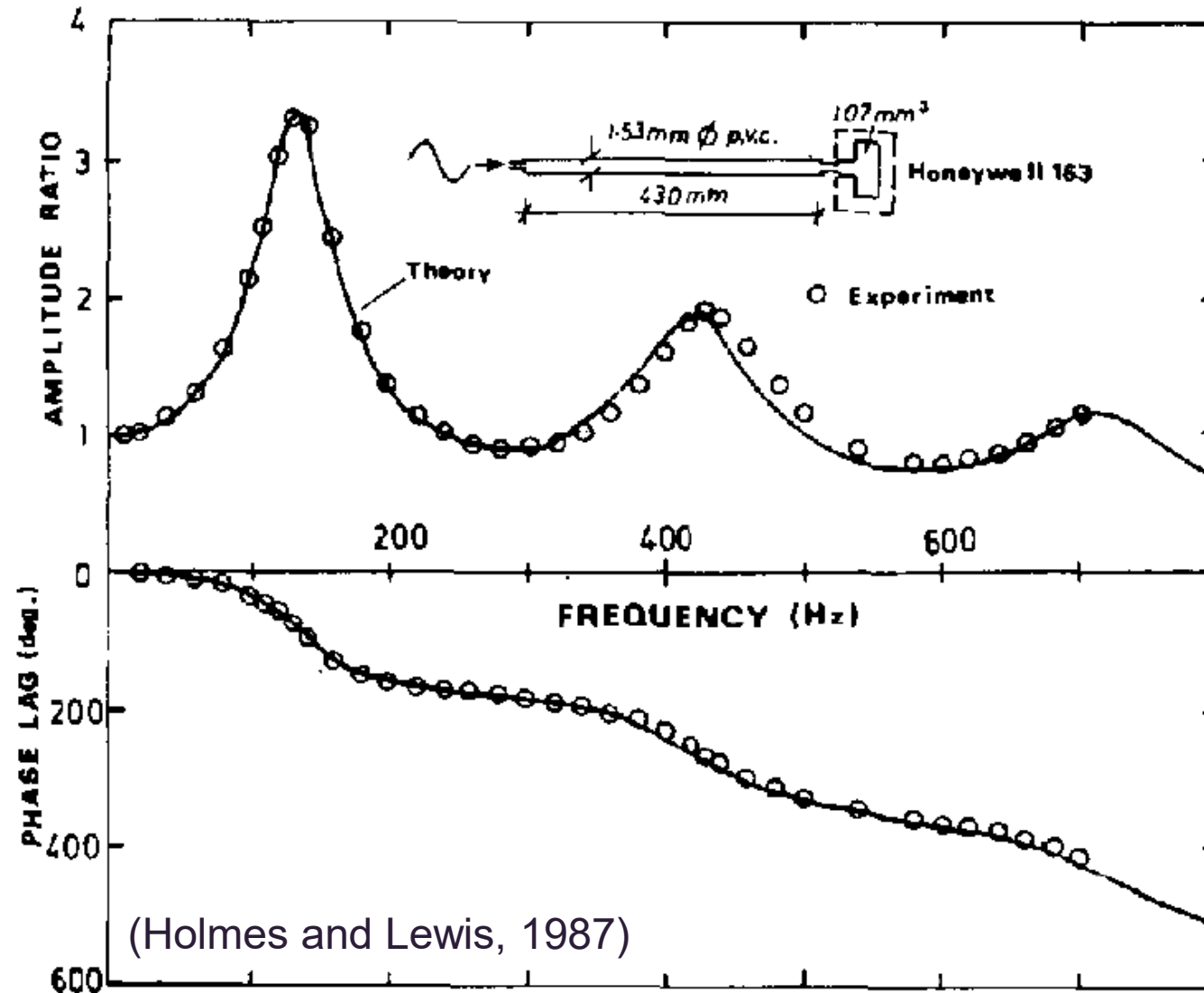


Pressure measurements

- Must ensure the frequency of pressure loading is the same as frequency response out of the measurement system.
- Dependent on:
 - Transducer volume
 - Tubing size
 - Tubing length
 - Restrictors
 - Scanning device



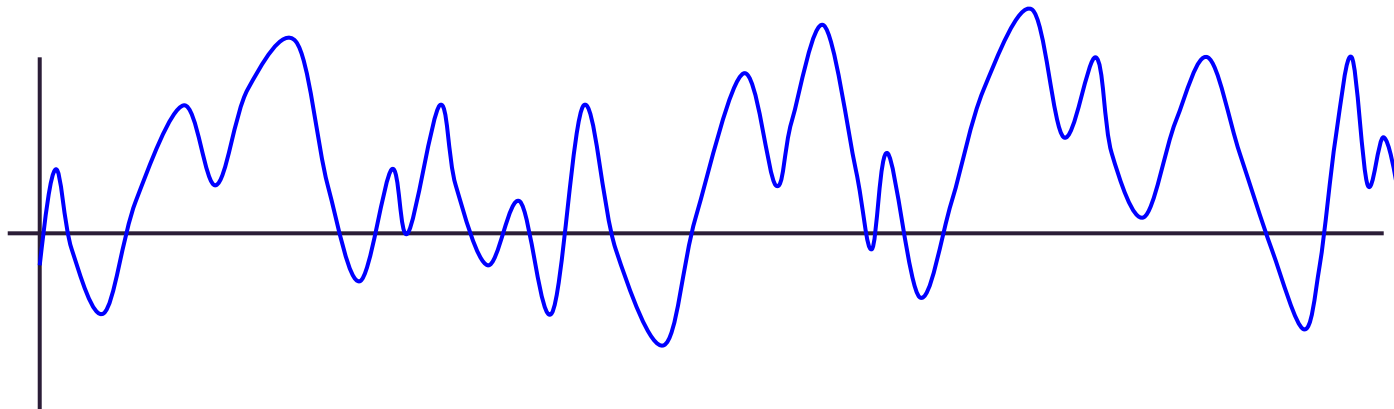
Pressure measurements



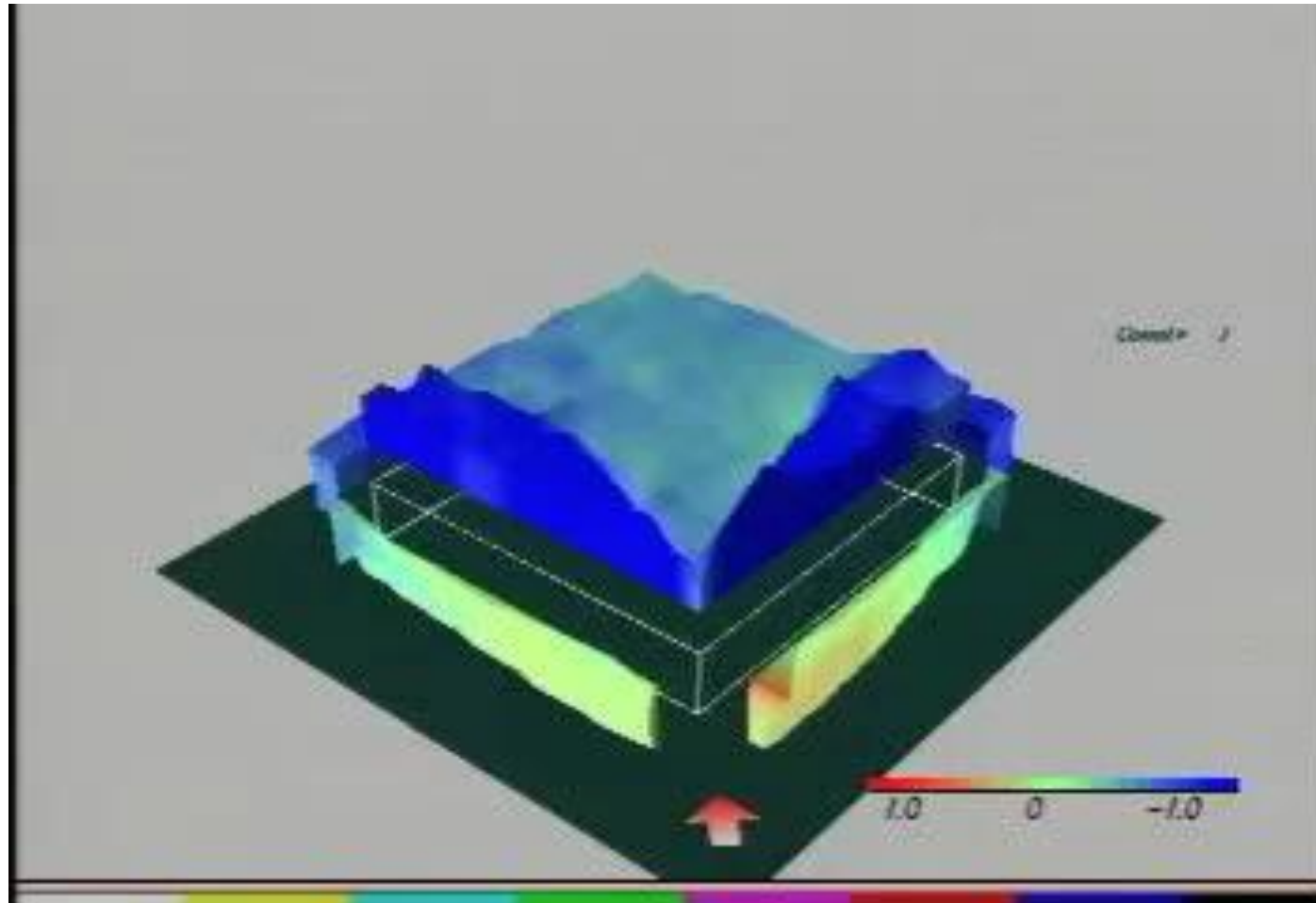
Pressure measurements

Determination of (statistical) peak

- Upcrossing
- Sampling of Maxima
- EVA on short period peaks
- Extremes of repeated experiments



Pressure measurements



Courtesy Y. Tamura

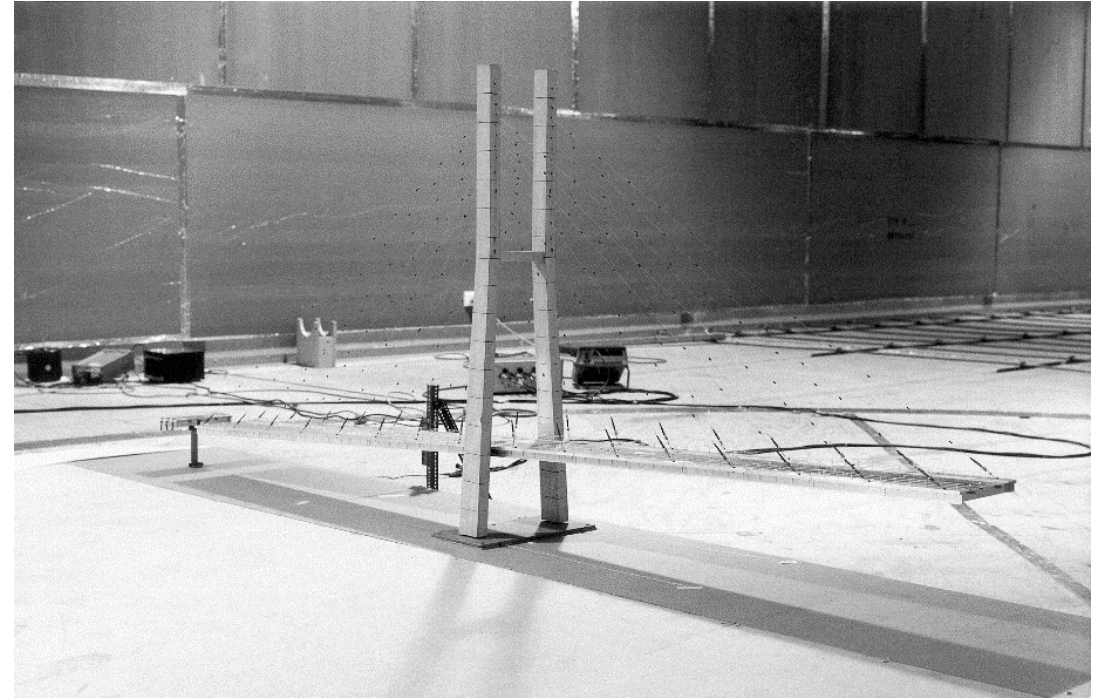
Overall load/response

Three primary methods used for buildings – special structures may require a fourth

1. Aeroelastic testing
2. High frequency base balance
3. Integrated pressures
4. Section testing

Aeroelastic tests

- Structural properties modelled in detail.
- Replica, spine, lumped mass.
- All physical scaling laws must be followed.
- All response results are obtained directly.
- Very expensive.



MOVIE

Section tests

- Used to test portions of structures.
- Large displacements – complex measurement required.
- Common for bridges.



Courtesy: Ting Yang

High frequency base balance

- Only shear and moments measured.
- Aeroelastic behaviour not modelled.
- System stiffness too high for wind excitation.
- Spectral analysis techniques used.
- Response theoretically calculated.



Courtesy CPP Sydney

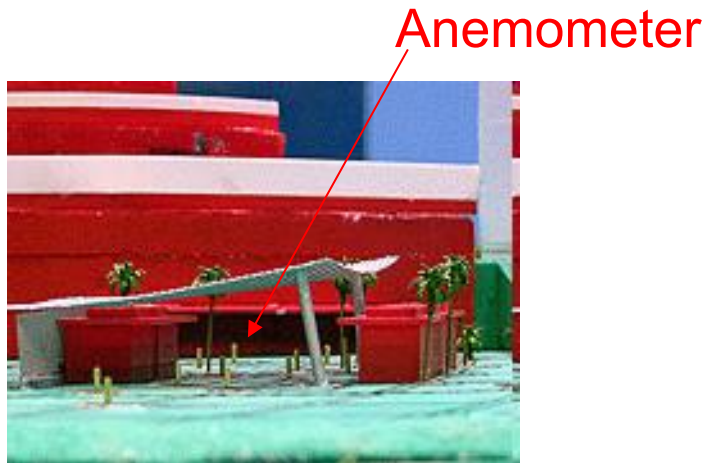
Integrated pressures

- Dense, simultaneous pressure measurement required.
- Tap numbers an issue.
- Must assign modal and area weighted factors for each tap.
- Dynamic analysis calculated.



Environmental comfort

- Compares measured wind speeds to perceivable discomfort or safety levels.
- Several techniques used to measure point locations



Summary

- An overview of the range of wind tunnels available for simulating boundaries.
- Seen how wind tunnels can be modified to accommodate different types of boundary layers (i.e. wind events)
- (Very) brief introduction to scaling requirements for wind tunnel testing.
- Brief discussion of the types of wind tunnel testing that can be done (from a building design/construction perspective).

Contact

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